Supplementary material

Budget-viable commitment for finite-language decoding under hard token budgets

Ya Bai and coauthors

S1 Historical motivation and implementation controls

The historical stopping study used Qwen2.5-7B-Instruct-1M and Mistral-7B-Instruct-v0.3 on GSM8K and ARC-Challenge (Yang et al., 2025; Mistral AI, 2024; Cobbe et al., 2021; Clark et al., 2018). Its frozen contract required the final line-start `####` answer to have a newline or EOS certificate and no later non-whitespace content. It did not validate an arbitrary surrounding document. Uniform ran to EOS or a 512-token cap; Complete returned the earliest certified terminal answer line. This is historical motivation, not a test of BVC.

Table S1: Historical Uniform-to-Complete paired validity repairs. Each model/task cell contains 600 questions.

Model	Task	n	Repairs	Correct	Wrong	Losses
Mistral	ARC-C	600	47	19	28	0
Mistral	GSM8K	600	129	81	48	0
Qwen	ARC-C	600	0	0	0	0
Qwen	GSM8K	600	28	14	14	0

Across 2,400 model–question pairs, 204 outputs changed from invalid to valid: 114 were correct and 90 were not. No validity loss was observed. Last-marked-answer rescoring gave zero Complete-minus-Uniform accuracy difference in all four cells; the closure-balanced heuristic produced a negative difference for Qwen/GSM8K. The old ten-table supplement remains in the repository and is retained rather than replaced by the newer task results.

The newly supplied historical raw archive contains 25,200 records: 480 pilot, 6,000 development, 14,400 main and 4,320 timing. The later three-GPU archive has 400 pilot

and 10,800 formal timing records plus 480 warmups. The offline audit reproduces strict main scoring and all five archived follow-up scoring contracts. The historical parser is frozen separately because its column-start marker and numeric syntax differ from the later prototype. Historical hardware/measurement boundaries are not pooled with the new studies.

The wrapper adapter binds a complete recognized candidate and attempts only supported closing suffixes. Its reserve is 16 tokens within the 128-token wrapper cap. Candidate recognition precedes a no-candidate reserve deadline at the same step; EOS before acceptance abstains. Unsupported states abstain rather than searching for a later candidate. BVC certifies 27/32 Qwen and 30/32 Mistral wrapper requests with no legal token pruned. Format-only controls compare deterministic and model-guided completion of the same bound answer, matching acceptance and value preservation when feasible and abstaining below the minimum.

S2 Follow-up trajectories and full results

Table S2: Original matched-prefix BVC/Grammar grid, $n = 96$ in every row. The original cap24 option-text correct-commit endpoint remains the primary family. Other policies and costs are retained in the released all-baseline CSV.

Model	Encoding	Cap	BVC cert.	G cert.	BVC correct	G correct
Qwen	Label	8	96	96	85	85
Qwen	Label	24	96	96	85	85
Qwen	Label	64	96	96	85	85
Qwen	Option text	8	84	69	54	54
Qwen	Option text	24	96	96	76	76
Qwen	Option text	64	96	96	76	76
Mistral	Label	8	96	96	66	66
Mistral	Label	24	96	96	66	66
Mistral	Label	64	96	96	66	66
Mistral	Option text	8	65	53	41	39
Mistral	Option text	24	96	96	66	66
Mistral	Option text	64	96	96	66	66

Each boundary condition reuses all 64 questions per model. The fixed8 condition is a global cap; exact-minimum and wide conditions use each case/model's minimum and

maximum canonical suffix lengths. They are structural controls, not three independent tasks or a continuous budget-response curve.

Table S3: Native Grammar successes preserved by BVC. Counts contain repeated questions across conditions.

Model	Condition	G cert.	Same tokens	Added cert.	Added correct	Lost
Qwen	Fixed 8	44	44	13	0	0
Qwen	Minimum	24	24	40	2	0
Qwen	Maximum	64	64	0	0	0
Mistral	Fixed 8	37	37	8	1	0
Mistral	Minimum	26	26	38	1	0
Mistral	Maximum	64	64	0	0	0

All 259 follow-up already-certified trajectories are retained; the original 96-question experiment retains 1,082 across its 12 model/encoding/cap cells. The stored returned IDs in the later fixed8 subset match the original for BVC, Grammar, XGrammar and Shortest for every model/question. These are consistency checks, not independent observations.

Table S4: All follow-up native endpoints and token accounting, $n = 64$ per row. Generated and inserted tokens are separate.

Model	Budget	Policy	Cert.	Correct	Wrong	Fwd	Gen.	Ins.
Qwen	Fixed 8	BVC	57	35	22	310	310	0
Qwen	Fixed 8	Grammar	44	35	9	378	378	0
Qwen	Fixed 8	Sequence score	57	34	23	57	0	308
Qwen	Fixed 8	XGrammar	39	31	8	417	417	0
Qwen	Fixed 8	Shortest	57	9	48	0	0	273
Qwen	Minimum	BVC	64	19	45	341	341	0
Qwen	Minimum	Grammar	24	17	7	341	341	0
Qwen	Minimum	Sequence score	64	19	45	64	0	341
Qwen	Minimum	XGrammar	0	0	0	341	341	0
Qwen	Minimum	Shortest	64	9	55	0	0	341
Qwen	Maximum	BVC	64	52	12	424	424	0
Qwen	Maximum	Grammar	64	52	12	424	424	0
Qwen	Maximum	Sequence score	64	50	14	64	0	421
Qwen	Maximum	XGrammar	31	25	6	455	455	0
Qwen	Maximum	Shortest	64	9	55	0	0	341
Mistral	Fixed 8	BVC	45	28	17	273	273	0
Mistral	Fixed 8	Grammar	37	27	10	427	427	0
Mistral	Fixed 8	Sequence score	45	27	18	45	0	274
Mistral	Fixed 8	XGrammar	37	27	10	427	427	0
Mistral	Fixed 8	Shortest	45	11	34	0	0	243
Mistral	Minimum	BVC	64	18	46	435	435	0
Mistral	Minimum	Grammar	26	17	9	435	435	0
Mistral	Minimum	Sequence score	64	17	47	64	0	435
Mistral	Minimum	XGrammar	26	17	9	435	435	0
Mistral	Minimum	Shortest	64	12	52	0	0	435

Model	Budget	Policy	Cert.	Correct	Wrong	Fwd	Gen.	Ins.
Mistral	Maximum	BVC	64	43	21	514	514	0
Mistral	Maximum	Grammar	64	43	21	514	514	0
Mistral	Maximum	Sequence score	64	43	21	64	0	512
Mistral	Maximum	XGrammar	63	43	20	514	514	0
Mistral	Maximum	Shortest	64	12	52	0	0	435

The accompanying CSV files also retain compiler time, failed-request cost, candidate suffix tokens and candidate-scoring input tokens. Candidate scoring uses one batch of feasible candidates, not one autoregressive forward per emitted token. The recorded call counts therefore do not establish proportional compute savings across these strategies.

Table S5: Exploratory accepted-answer stability against certified Grammar outputs. Retention is same certified answer/G certified; regression and correction use offline gold only after inference.

Model	Budget	Policy	Retention	Lost correct	Corrected wrong
Qwen	Fixed 8	BVC	44/44	0	0
Qwen	Fixed 8	Sequence score	41/44	2	1
Qwen	Fixed 8	Shortest	9/44	29	3
Qwen	Fixed 8	XGrammar	39/44	4	0
Qwen	Minimum	BVC	24/24	0	0
Qwen	Minimum	Sequence score	24/24	0	0
Qwen	Minimum	Shortest	10/24	11	0
Qwen	Minimum	XGrammar	0/24	17	0
Qwen	Maximum	BVC	64/64	0	0
Qwen	Maximum	Sequence score	58/64	4	2
Qwen	Maximum	Shortest	10/64	46	3
Qwen	Maximum	XGrammar	31/64	27	0
Mistral	Fixed 8	BVC	37/37	0	0
Mistral	Fixed 8	Sequence score	35/37	1	0
Mistral	Fixed 8	Shortest	14/37	17	1
Mistral	Fixed 8	XGrammar	37/37	0	0
Mistral	Minimum	BVC	26/26	0	0
Mistral	Minimum	Sequence score	24/26	1	0
Mistral	Minimum	Shortest	17/26	6	0
Mistral	Minimum	XGrammar	26/26	0	0
Mistral	Maximum	BVC	64/64	0	0
Mistral	Maximum	Sequence score	59/64	2	2
Mistral	Maximum	Shortest	17/64	32	1
Mistral	Maximum	XGrammar	63/64	0	0

The diagnostic preserves the original endpoint and raw statuses. BVC's trajectory theorem implies complete retention under its assumptions; other policies optimize different

objectives or use different tokenizations and termination. The comparison therefore measures accepted-answer stability, not overall algorithm superiority. Some alternative changes improve correctness. All repeated-question dependence and exploratory qualifications still apply.

S3 Reference and terminal diagnostics

At fixed8, all 13 additional Qwen and seven of eight additional Mistral native completions have over-budget canonical reference paths. The per-request table `rescued_cases.csv` gives IDs, paired request IDs, reference lengths, budgets, dead-entry steps and final outputs. Gold is used only by offline scoring. The main manuscript retains all-request endpoints rather than substituting a favorable reference-reachable subset.

Table S6: All native versus weaker text-only endpoints. Outcome cells give count/correct count; the final column is text-valid but not committed.

Model	Budget	Policy	Native	Text-only	Uncommitted
Qwen	Fixed 8	BVC	57/35	57/35	0
Qwen	Fixed 8	Grammar	44/35	44/35	0
Qwen	Fixed 8	XGrammar	39/31	44/35	5
Qwen	Minimum	BVC	64/19	64/19	0
Qwen	Minimum	Grammar	24/17	24/17	0
Qwen	Minimum	XGrammar	0/0	24/17	24
Qwen	Maximum	BVC	64/52	64/52	0
Qwen	Maximum	Grammar	64/52	64/52	0
Qwen	Maximum	XGrammar	31/25	64/52	33
Mistral	Fixed 8	BVC	45/28	45/28	0
Mistral	Fixed 8	Grammar	37/27	43/30	6
Mistral	Fixed 8	XGrammar	37/27	42/30	5
Mistral	Minimum	BVC	64/18	64/18	0
Mistral	Minimum	Grammar	26/17	42/28	16
Mistral	Minimum	XGrammar	26/17	42/28	16
Mistral	Maximum	BVC	64/43	64/43	0
Mistral	Maximum	Grammar	64/43	64/43	0
Mistral	Maximum	XGrammar	63/43	64/43	1

There are 106 consumer-valid but uncommitted records across all follow-up policies and repeated budget conditions. This is endpoint sensitivity, not raw corruption or new executed returns. A grammar trajectory can be canonically non-completable but already satisfy the weaker text predicate. Correctness or utility guarantees against a weaker early-return comparator do not follow from native-trajectory preservation.

S4 Wave-stratified descriptive timing

The original timing phase uses two sequential waves, with one Qwen and one Mistral worker per wave on distinct A800 GPUs after a resident-model barrier. Each model/wave/policy cell has 24 requests on eight questions, each repeated three times. Model/GPU identity and question shards are coupled. The two waves do not identify a wave random effect. The follow-up single-GPU measurements include correctness instrumentation; historical three-GPU measurements use a different regime.

Table S7: Original supplied-prefix mean request time in milliseconds, by wave and policy. Means are descriptive and are not complete upstream-generation costs.

Wave	Model	Policy	Requests	Mean ms
0	Mistral	BVC	24	150.33
0	Mistral	Grammar	24	150.40
0	Mistral	Sequence score	24	62.70
0	Mistral	Uniform	24	149.59
0	Mistral	XGrammar	24	153.03
0	Qwen	BVC	24	120.05
0	Qwen	Grammar	24	119.99
0	Qwen	Sequence score	24	60.77
0	Qwen	Uniform	24	167.60
0	Qwen	XGrammar	24	147.45
1	Mistral	BVC	24	220.49
1	Mistral	Grammar	24	221.53
1	Mistral	Sequence score	24	62.67
1	Mistral	Uniform	24	241.28
1	Mistral	XGrammar	24	222.42
1	Qwen	BVC	24	164.49
1	Qwen	Grammar	24	163.66
1	Qwen	Sequence score	24	62.21
1	Qwen	Uniform	24	239.53
1	Qwen	XGrammar	24	189.51

All fixed8 follow-up and original matched-prefix baseline fields are released as CSV. No timing superiority over candidate scoring is inferred, and no timing interval is newly claimed for the historical concurrent archive. Failed attempts are retained in accounting.

S5 Statistical and reproducibility details

The original primary endpoint is all-request correct-commit yield in option-text cap24, assessed separately for two models. Its conservative per-cell 97.5% intervals account for the two-cell family. The follow-up fixed8 family uses feasible-subset certification, again

with two model cells; its exploratory, reused-data status remains explicit. Other intervals are descriptive and unadjusted across additional settings.

For w paired wins, l paired losses and n questions, simultaneous exact binomial bounds for win and loss probabilities use tail probability $\alpha/4$ for each bound. Subtracting the loss bounds from the win bounds yields the reported conservative paired $1 - \alpha$ interval. This construction has nonzero width under zero observed discordance. It must not be replaced by a degenerate bootstrap interval or interpreted as proof of equivalence. Question pairing is retained, and model/budget repetitions do not increase the number of independent questions.

The raw archive consists of the four dated study folders. Use each raw source once: original aggregate files are byte-identical to their wave copies; existing analysis/scored JSONL files are derived, not extra requests. All 443 files listed in the provided checksum manifest passed verification. Single-study path traces are reconstructed by direct enumeration of frozen paths, independently of stored runtime distance events. This does not independently retokenize absent model files or establish complete coverage of every legal text/tokenization.

Recorded model revisions are Qwen `e28526f7bb80e2a9c8af03b831a9af3812f18fba` and Mistral `c170c708c41dac9275d15a8fff4eca08d52bab71`. Hashes of metadata and weight files agree across recorded manifests; weights themselves are absent from the bundle. Both actual schedulers and matching inference sources are provided. No shutdown, remote-control or infrastructure script is executed by the offline audit.

The public repository is `https://github.com/Eov-firefly/learningclass`. The outer raw bundle SHA-256 is

`91fed3db13cca2b2d8ab65d5c6e0a714c0ad55626f5133a93f8be119e6e3782e`

The single-GPU and two-wave inner ZIP hashes are, respectively,

`874d09359bd953712b801b3ce40fa249da6646eb8041deda19acdfe8aa63aef1`

`356984d29742605b26261bb7cc73cce575f53c7e1687c04404ab3224760ef7bf`

The primary raw bundle is the dated raw ZIP at the repository root; its four folder names and nested paths are listed in the bundle README. The scripts and tables are under

`research_outputs/raw_audit_20261102`. Reproduce the audit from the repository root after extracting the archive with

```
audit_raw.py -bundle EXTRACTED_BUNDLE -out NEW_OUTPUT
```

using the script's full repository path and Python. The audit requires numpy/scipy, not model weights or GPU inference. The historical parser snapshot records its exact origin and hash; it retains the archived column-start marker and numeric grammar. All exact command lines and machine-readable verification records accompany the repository's audit README.

References

Peter Clark, Isaac Cowhey, Oren Etzioni, Tushar Khot, Ashish Sabharwal, Carissa Schoenick, and Oyvind Tafjord. Think you have solved question answering? Try ARC, the AI2 reasoning challenge. *arXiv preprint arXiv:1803.05457*, 2018. URL `https://arxiv.org/abs/1803.05457`.

Karl Cobbe, Vineet Kosaraju, Mohammad Bavarian, Mark Chen, Heewoo Jun, Lukasz Kaiser, Matthias Plappert, Jerry Tworek, Jacob Hilton, Reiichiro Nakano, Christopher Hesse, and John Schulman. Training verifiers to solve math word problems. *arXiv preprint arXiv:2110.14168*, 2021. URL `https://arxiv.org/abs/2110.14168`.

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